

Assessing and Predicting Federal Grant Cuts in North Carolina and UNC

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1. Introduction

Federal financial assistance awards fund research, education, infrastructure, and essential services across the United States. In 2024 alone, these programs totaled over \$1.1 trillion, reshaping local services, research agendas, and community development. When the Department of Government Efficiency (DOGE) began large-scale funding reductions in 2025, the consequences were felt not just nationally but directly at UNC, affecting students and faculty members. After reviewing federal assistance data and DOGE's cuts, our group asked two questions.

Our first question explores which characteristics of federal grants are most associated with budget cuts in North Carolina, and how accurately these cuts can be predicted from available data. We are looking specifically within North Carolina to determine effects on a more local level. This question matters because if cuts follow systematic patterns, state agencies and universities could anticipate losses and prepare contingency plans rather than react after funding disappears. Our findings could also be applied to future potential budget cuts under the current federal administration, which is still determined to cut down on federal spending even after the dissolution of DOGE last month.

Our second question asks which types of projects within the UNC system are most at risk of experiencing substantial federal cuts. We examined the entire UNC system, including UNC-Chapel Hill and other UNC institutions, because UNC-CH alone did not provide enough observations to train predictive models. This narrows our analysis to impacts on education and research across North Carolina's public universities. UNC-CH specifically received the highest level of federal funding in the state (about \$2.5 billion over the past five years), and based on DOGE's methodology, more than \$100 million was cut. Identifying which programs are most vulnerable could help campus administrators prioritize funding requests, seek alternative funding sources, and protect critical research activities.

2. Description of Data

We constructed our dataset from three sources to answer our questions. The first source is USAspending.gov, the federal government's system for reporting details about assistance awards. Federal agencies report information about each award, including how much money is obligated (*Total Obligation*), the agency responsible (*Agency*), the Catalog Federal Domestic Assistance program the award belongs to (*CFDA Title*), the recipient (*Recipient*), the state where the award is performed (*State*), and the fiscal year it was recorded (*FY*). We filtered our dataset to encompass grants affecting North Carolina from 2020

through 2025. Each row represents a single federal award, and the data were collected directly by federal agencies and grant recipients as required by law.

The second dataset comes from DOGE’s website, which lists the “savings” achieved when DOGE reduces or cancels awards. The DOGE dataset contains two important variables: the total possible expenditure for the project (*Value*) and the amount of spending that was avoided (*Savings*), which is the difference between that total value and the amount currently obligated (*Total Obligation*). An obligation refers to what the government is legally committed to pay for the grant. This number shows how much money DOGE claims to save by cancelling the grant, making *Savings* our target variable for the analysis.

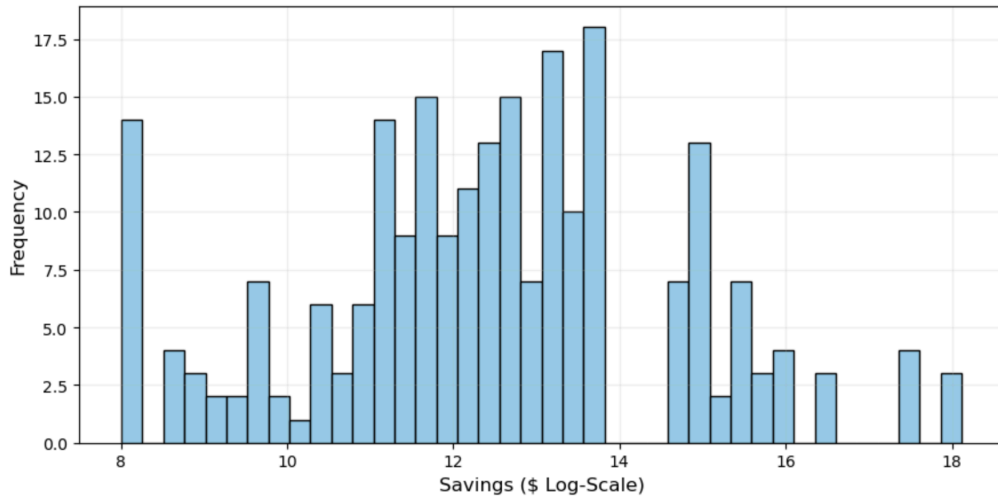
To merge the USAspending and DOGE datasets, we use the *FAIN* (Federal Award Identification Number), which is a unique ID assigned to each award by the federal government. The *FAIN* appears in both datasets, so it allows us to match detailed spending information to DOGE’s savings data. We drop this column after merging. We also created a new column that assigns each award to one of 6 broader policy domains (*Sector*) based on *CFDA Title*. Our final merged dataset shows information about grants that were cut in NC and contains 234 observations and 9 variables. Each row represents one grant, containing monetary information, program descriptions, agency names, and recipient information.

Table 1: Sample of Merged Dataset

Value	Savings	Total Obligation	Recipient	Agency	CFDA Title	State	FY	Sector
34998729.0	12993908.0	22004821.0	NaN	USAID	USAID FOREIGN ASSISTANCE F...	NORTH CAROLINA	2024.0	International Development
34998729.0	12993908.0	22004821.0	NaN	USAID	USAID FOREIGN ASSISTANCE F...	NORTH CAROLINA	2024.0	International Development
34722274.0	9273274.0	25449000.0	FAMILY HEALTH INTERNATIONAL	USAID	USAID FOREIGN ASSISTANCE F...	NORTH CAROLINA	2020.0	International Development
34722274.0	9273274.0	25449000.0	FAMILY HEALTH INTERNATIONAL	USAID	USAID FOREIGN ASSISTANCE F...	NORTH CAROLINA	2021.0	International Development
140000000.0	73943882.0	66056118.0	FAMILY HEALTH INTERNATIONAL	USAID	USAID FOREIGN ASSISTANCE F...	NORTH CAROLINA	2022.0	International Development

The cleaning process involved removing observations with missing *Savings*, converting monetary fields to numeric types, and scaling down large values for easier interpretation and visualization. Because both research questions rely on *Savings* as the outcome variable, we first explored its distribution. *Savings* range from roughly \$3,000 to \$66 million, indicating substantial variation in the size of DOGE’s reductions. When plotted on a log scale, the distribution becomes much smoother and approximately normal, centered between about \$450,000 and \$1.2 million. This helps visualize savings across projects without extreme values dominating the plot.

Figure 1: Distribution of Savings in Log Scale



3. Methodology and Results

Question 1

The first goal of our analysis is to understand which factors are associated with DOGE's *Savings* and whether we can predict these with models. We ran correlation tests between *Savings* and all variables to determine which ones may be significant in an analysis. We focus on three main predictors: *Value*, *Total Obligation*, and *Sector*. We treat *Savings* as the target variable, measured in millions of dollars. We split the data into a training set and a test set using a 70/30 ratio and fit 4 models: linear regression, ridge regression, a decision tree, and a random forest. We evaluate each model using out-of-sample R-squared (OSR^2), out-of-sample mean squared error (MSE), and out-of-sample mean absolute error (MAE).

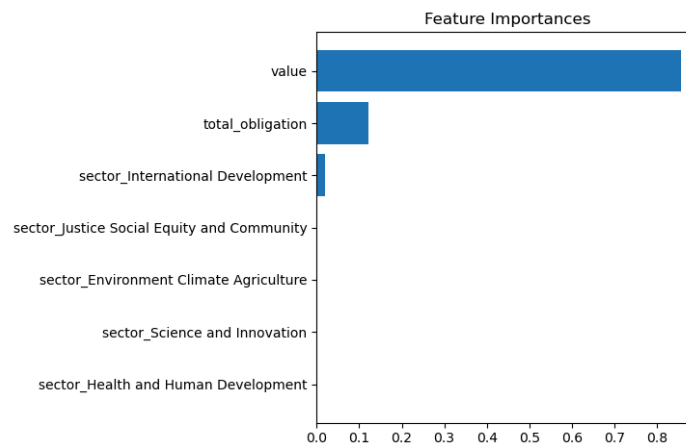
The linear regression model provides a helpful baseline. It assumes a straight-line relationship between *Savings* and the predictors. This model shows a large share of the variation in savings and achieves an OSR^2 of about 0.893 on the test set. Its mean absolute error is around 1.177 million dollars. The ridge regression model, which uses an L2 penalty to stabilize coefficients when predictors are correlated, performs slightly better. These results show that *Value* and *Total Obligation* account for a substantial share of predictive power and that even a simple model can capture much of the pattern. Next, we test a decision tree to allow for non-linear relationships. The tree sets a minimum of five samples in each leaf and uses a cost-complexity pruning parameter to avoid overfitting. Even with these settings, the single tree does not generalize well. Its OSR^2 is around 0.75, much lower than the linear models. This suggests that a single tree is too unstable or too simple for this prediction task. The random forest model performs best. With five hundred trees, it captures complex interactions between *Value*, *Total Obligation*, and *Sector*. On the test set, the random forest reaches an OSR^2 of about 0.95 and has a mean absolute error of under one million dollars. In other words, the model can predict DOGE savings with high accuracy across a wide range of values. This result appears visually in the plot comparing actual and predicted savings, where most points line up close to the diagonal line that represents perfect prediction.

Table 2: Comparison Table to Evaluate Model Efficacy

	Linear Regression	Ridge Regression	Decision Tree Regressor	Random Forest
OSR2	0.893	0.895	0.750	0.950
Out-of-sample MSE	19.4547	19.1190	45.3895	9.1371
Out-of-sample MAE	1.177	1.150	1.544	0.871

To answer our first question, we examined which variables are most important for predicting DOGE’s cuts. The feature-importance results show that *Value* is by far the strongest predictor, followed by *Total Obligation*, with several sector categories contributing smaller but still meaningful effects. International development programs, particularly those associated with USAID, appear more likely to face substantial reductions, which is consistent with the raw data. A notable finding is that *Value* has higher correlation and feature importance than *Total Obligation*. This suggests that DOGE’s cutting decisions may be based on the potential expenditure (the ceiling amount the government could spend) rather than the actual obligation (the amount the government is legally committing to pay). This is important given criticisms of DOGE’s methodology for overstating “savings” by relying on ceiling values instead of realistic spending estimates. Nonetheless, this pattern provides a strong predictive signal for modeling purposes.

Figure 2: Feature Importances (Derived from Random Forest Model)



Question 2

Our second goal is to determine which types of UNC projects are at high risk of experiencing large funding cuts. We began by filtering the dataset to observations where the *Recipient* is the University of North Carolina. We created a binary outcome variable that labels each project as high risk ($y=1$) if its

savings are above the median savings value for UNC projects and low risk ($y=0$) otherwise. We used all features (excluding *Savings*) and preprocessed categorical data by using one-hot encoding.

To build the predictive model, we constructed a pipeline in scikit-learn and evaluated 4 classifiers: logistic regression, decision tree, random forest, and gradient boosting. For each model, we performed 5-fold cross-validation over multiple hyperparameter settings and measured performance using the ROC-AUC metric. Hyperparameters were selected based on the highest mean AUC across folds, which ensures that parameter choices reflect out-of-sample performance rather than a single train-test split. Across all four algorithms, Gradient Boosting achieved the highest mean AUC and smallest standard deviation, indicating it offers the most accurate and stable predictions for this problem.

Table 3: Comparison Table to Evaluate Model Efficacy

Model	Best Parameters	Mean AUC	Std AUC
Logistic Regression	{'C': 0.01}	0.795486	0.262203
Decision Tree	{'max_depth': 2}	0.811111	0.239373
Random Forest	{'max_depth': 3, 'n_estimators': 100}	0.836111	0.235276
Gradient Boosting	{'max_depth': 2, 'learning_rate': 0.05}	0.852778	0.145496

After selecting gradient boosting as the best-performing model, we refit it using the optimal parameters and estimated the probability that each UNC award falls into the high-risk category. We then summarized these probabilities by *CFDA Title* to identify which areas face the greatest potential cuts. The results show that UNC’s Teacher Quality Partnership grants (4 awards) and STEM Education programs (9 awards) have the highest predicted risk, indicating these program areas may be more vulnerable to further funding reductions. Although this analysis is based on the current dataset, the modeling approach provides a practical tool that can be applied to future UNC awards to flag vulnerable projects and support proactive planning.

Table 4: Applied Summary of Predicted Risk Across CFDA Titles

CFDA Title	Number_of_Awards	Mean_Predicted_Risk	Maximum_Predicted_Risk
NSF TECHNOLOGY, INNOVATION, AND PARTNERSHIPS	1	0.969099	0.969099
TEACHER QUALITY PARTNERSHIP GRANTS	4	0.964138	0.971159
OFFICE OF RESEARCH AND DEVELOPMENT CONSOLIDATED RESEARCH/TRAINING/FELLOWSHIPS	1	0.963319	0.969099
STEM EDUCATION (FORMERLY EDUCATION AND HUMAN RESOURCES)	9	0.799015	0.980445
COMPUTER AND INFORMATION SCIENCE AND ENGINEERING	2	0.464034	0.787830
EDUCATION AND HUMAN RESOURCES	13	0.351334	0.965623
SCIENCE TO ACHIEVE RESULTS (STAR) RESEARCH PROGRAM	2	0.161546	0.239309
SOCIAL, BEHAVIORAL, AND ECONOMIC SCIENCES	6	0.159188	0.685752
CITIZENSHIP EDUCATION AND TRAINING	1	0.055482	0.055482
BASIC SCIENTIFIC RESEARCH	1	0.042792	0.042792

4. Conclusion

To summarize, our analysis shows that the total possible expenditure (*Value*) and the total amount legally committed by the government (*Total Obligation*) are the strongest predictors of DOGE savings, with international development programs disproportionately affected. Within the UNC system, the highest predicted risks occur in Teacher Quality Partnership and STEM Education grants. These findings matter for universities and state agencies that rely heavily on federal support. If reductions can be predicted from publicly available data, institutions can anticipate losses, plan alternative funding strategies, and protect essential programs before cuts materialize. Our models also provide a foundation that universities and organizations could adapt to forecast future savings under similar federal policies. Between the DOGE cuts and an extensive government shutdown in 2025, this could be very topical under the current administration that federally funded institutions should consider in allocating funds.

Our analysis also highlights an unusual aspect: *Value* was more predictive than *Total Obligation*, implying that DOGE decisions may be tied more to potential spending ceilings than actual commitments. This result aligns with public critiques of inflated savings estimates. Understanding this pattern helps stakeholders interpret DOGE's calculations and evaluate whether reported "savings" reflect meaningful efficiency or simply bookkeeping assumptions.

Future work could expand in several directions. One recurring question in our group was whether DOGE savings exhibit regional or geographic patterns (e.g. across rurality, urbanity, etc.). Incorporating geographic covariates such as rural–urban classification codes, county-level socioeconomic indicators, or regional funding baselines could help determine whether location systematically influences reductions. This question is especially relevant within North Carolina. For example, UNC–Chapel Hill is suburban, whereas Appalachian State serves a more rural region. Comparing predicted risks between these campuses could reveal whether rural institutions face more severe savings pressure, whether suburban research universities are targeted differently, or whether DOGE's decisions are geographically neutral. Understanding such dynamics would help universities across the state (and the country) anticipate vulnerabilities tied to their regional context and plan accordingly.

Our analysis found DOGE's savings were modeled in a predictable manner that directly impacts public institutions like universities, which marks a pivot towards economic security. Now than ever, it is more important for federally funded organizations to get ahead of budgetary threats.